Ethanol Blending in Gasoline – Taiwan

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U.S. GRAINS & BIOPRODUCTS COUNCIL

20 F St. NW, Suite 900 \ Washington, DC 20001
202-789-0789 \ www.grains.org

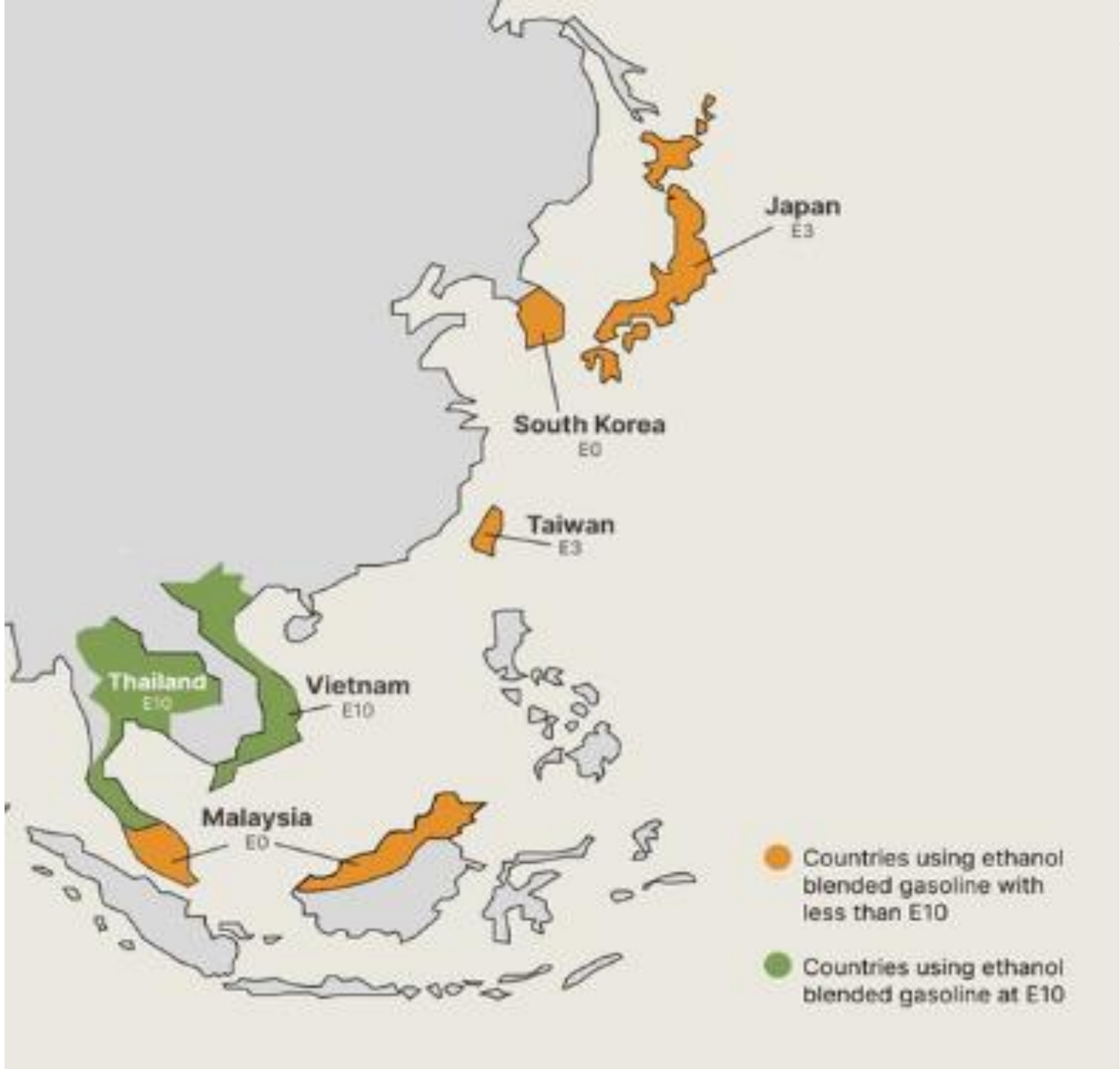
FARO90

www.faro90.com
+52 55 6122 2255

Ethanol Blending in Asia

There are important fuel quality and environmental impact of vehicle emission challenges in these regions.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Asia to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.



Seven countries with potential and additional use of ethanol were studied: 1) Gasoline market profiles; 2) Optimization of gasoline blends with ethanol and, 3) Environmental impact of gasolines blended with ethanol.

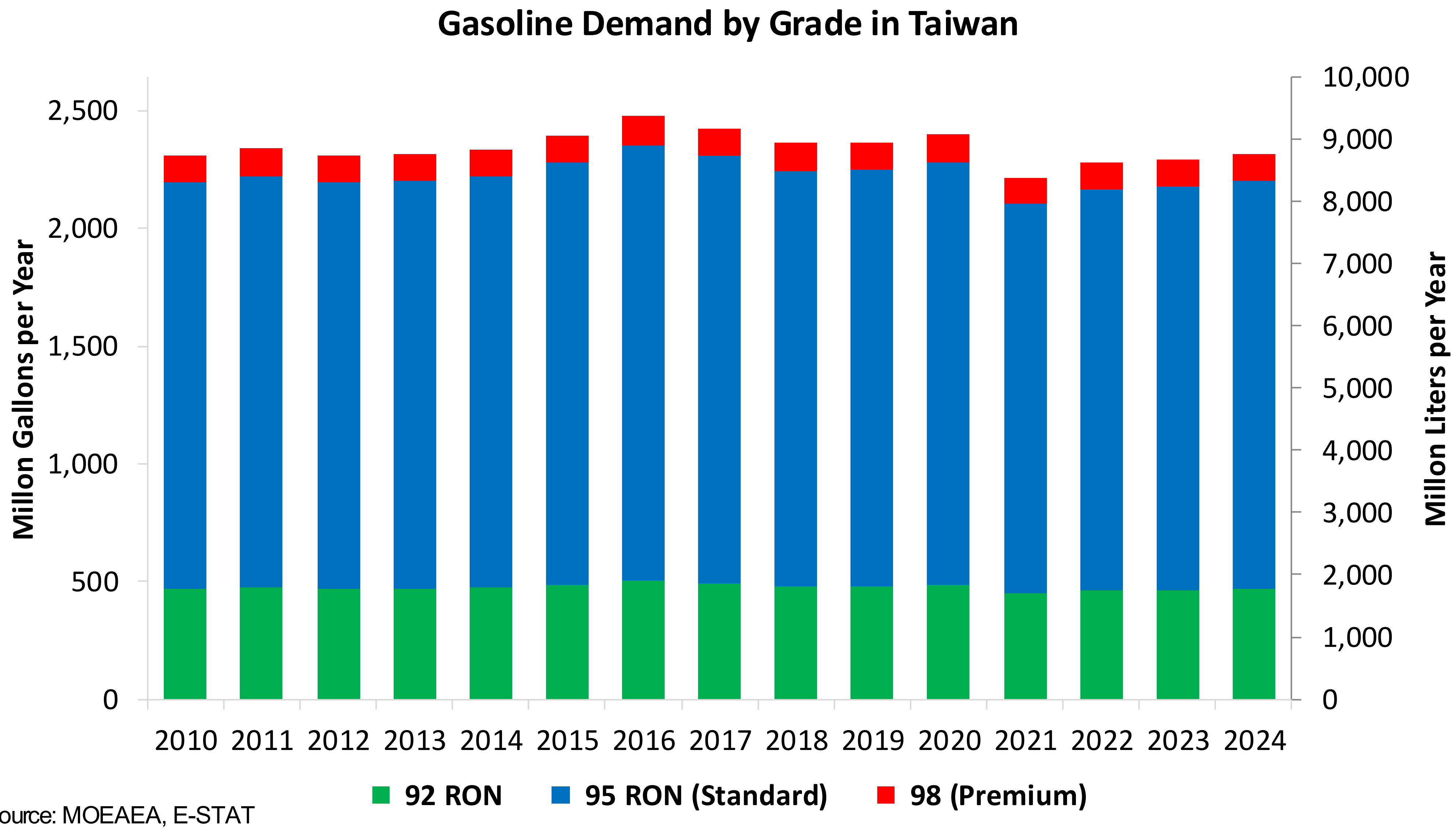
Ethanol Gasoline Blending in Taiwan

In 2024, gasoline consumption in Taiwan was 8,761 million liters (2,314 million gallons) and growth rate was -0.4%. Gasoline production and net imports were 12,450 and -3,689 million liters (3,289 and -975 million gallons) correspondingly. Taiwan offers three main grades: RON 92, RON 95 and RON 98) with an average octane of 94.5 RON.

Taiwan has policies to promote bioethanol in transportation to reduce imported fuel dependency and lower carbon emissions, using it as an oxygenate additive. While there was a limited trial of E3 gasohol in the mid-2000s, the lack of domestic production and insufficient incentives led to its discontinuation. Only a few gas stations offer it. Ethanol is supplied by imports and corresponds to an average content of 0.2%. CPC's 2026 budget indicates that the company plans to procure 96,000 liters of ethanol for E3 blends.
Future bioethanol use depends on developing profitable cellulosic ethanol production from non-food waste like rice straw and food waste, rather than imported feedstocks.

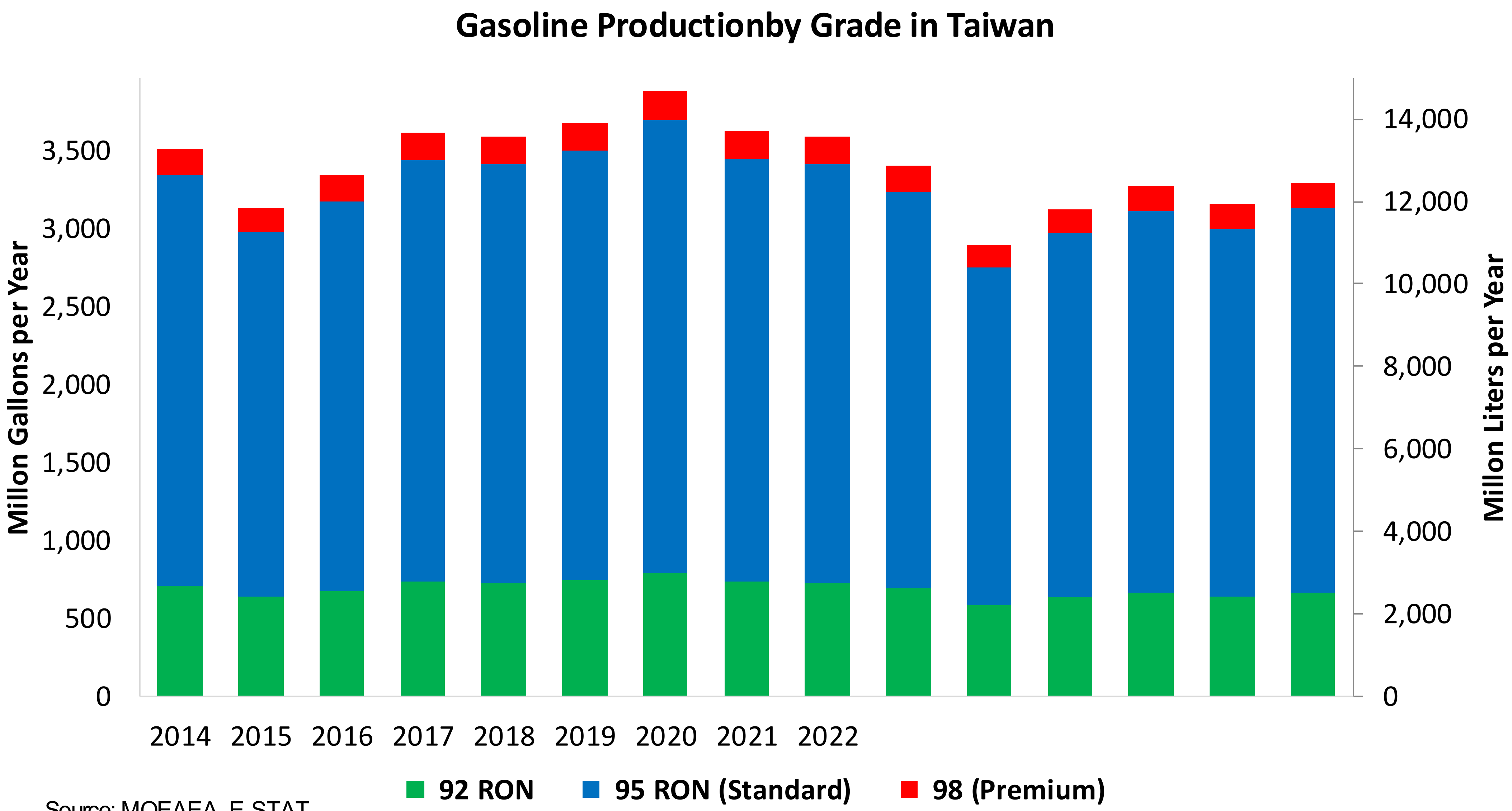
Source: MOEAEA, E-STAT, MOENV, USDA

Gasoline Demand in Taiwan



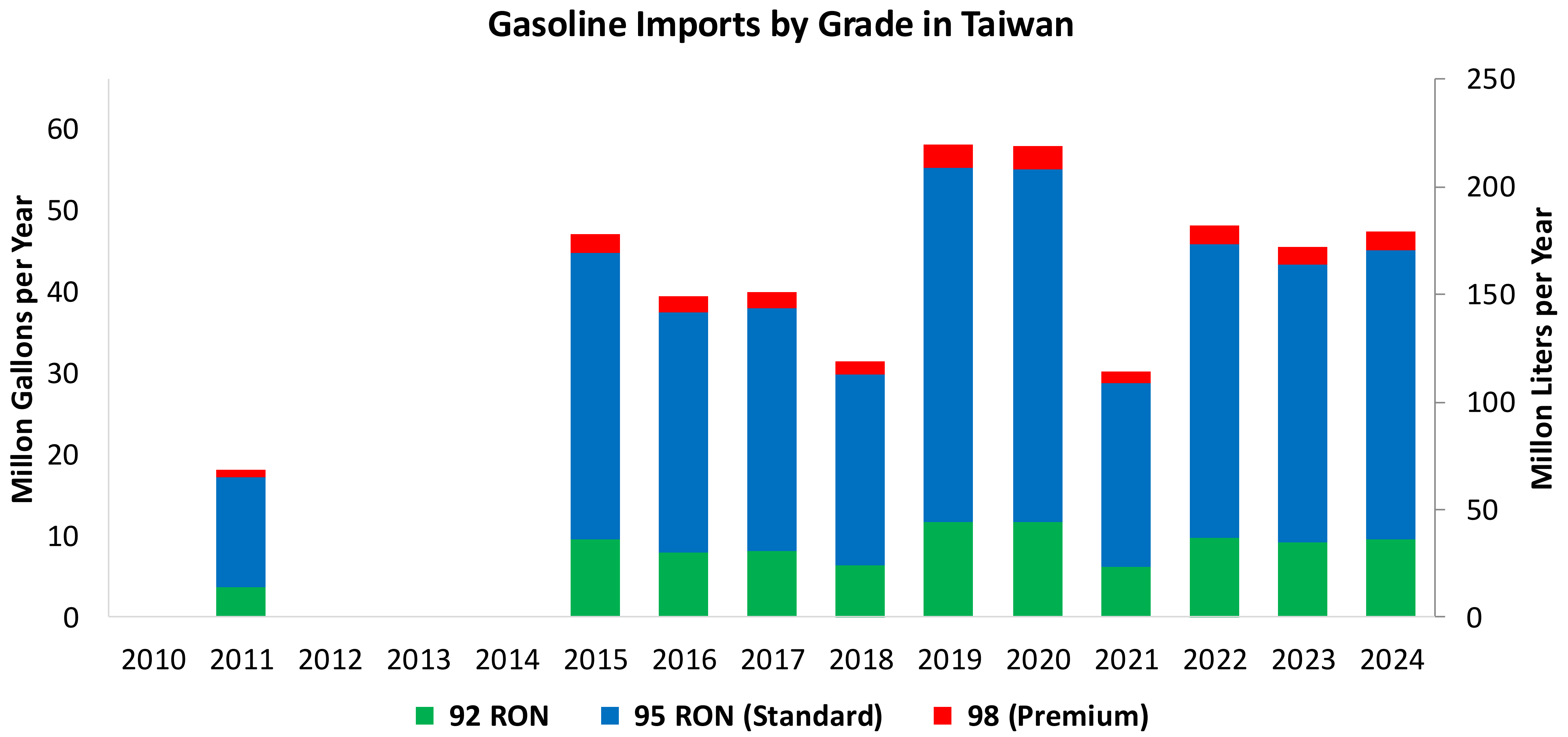
Source: MOEAEA, E-STAT

Gasoline Production in Taiwan



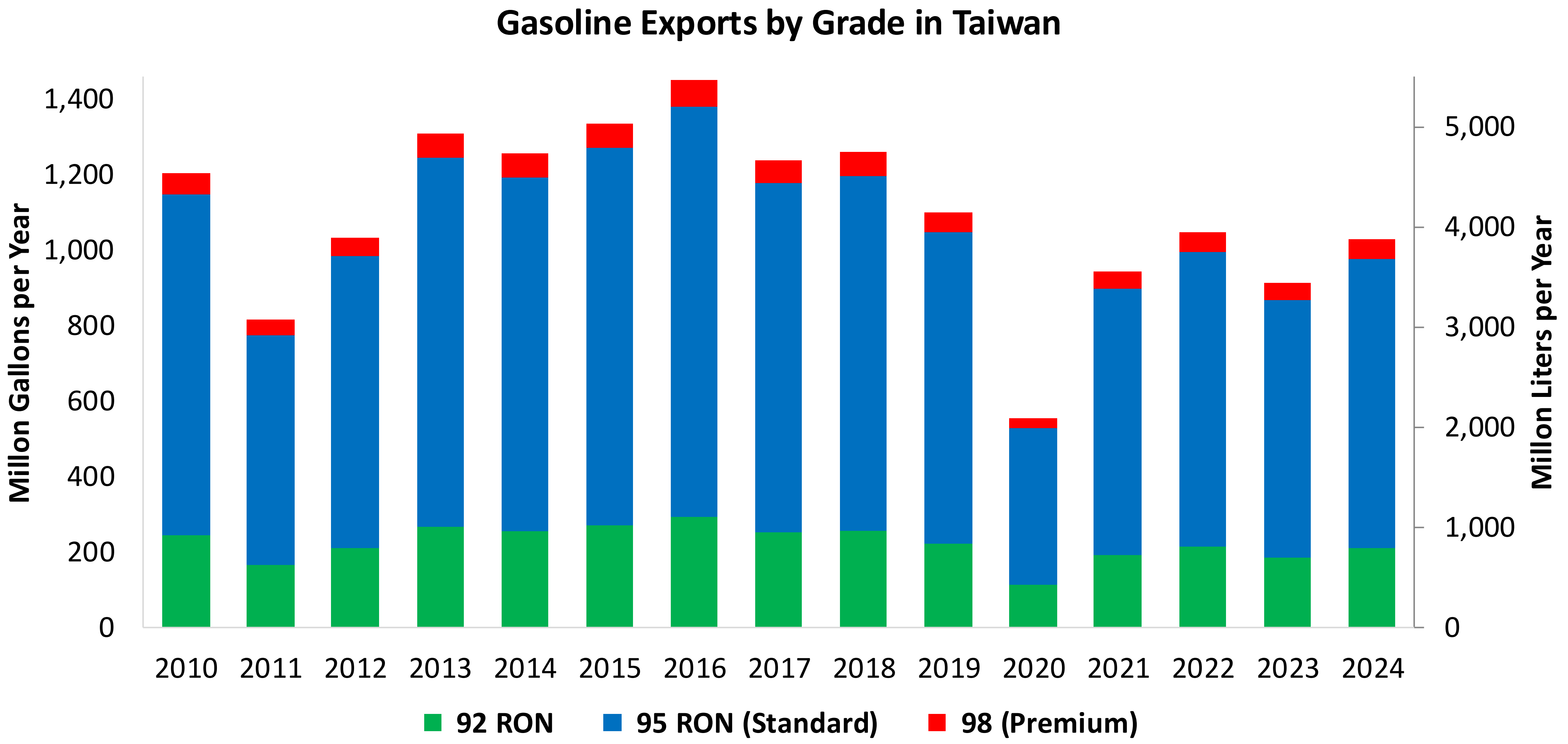
Source: MOEAEA, E-STAT

Gasoline Imports in Taiwan



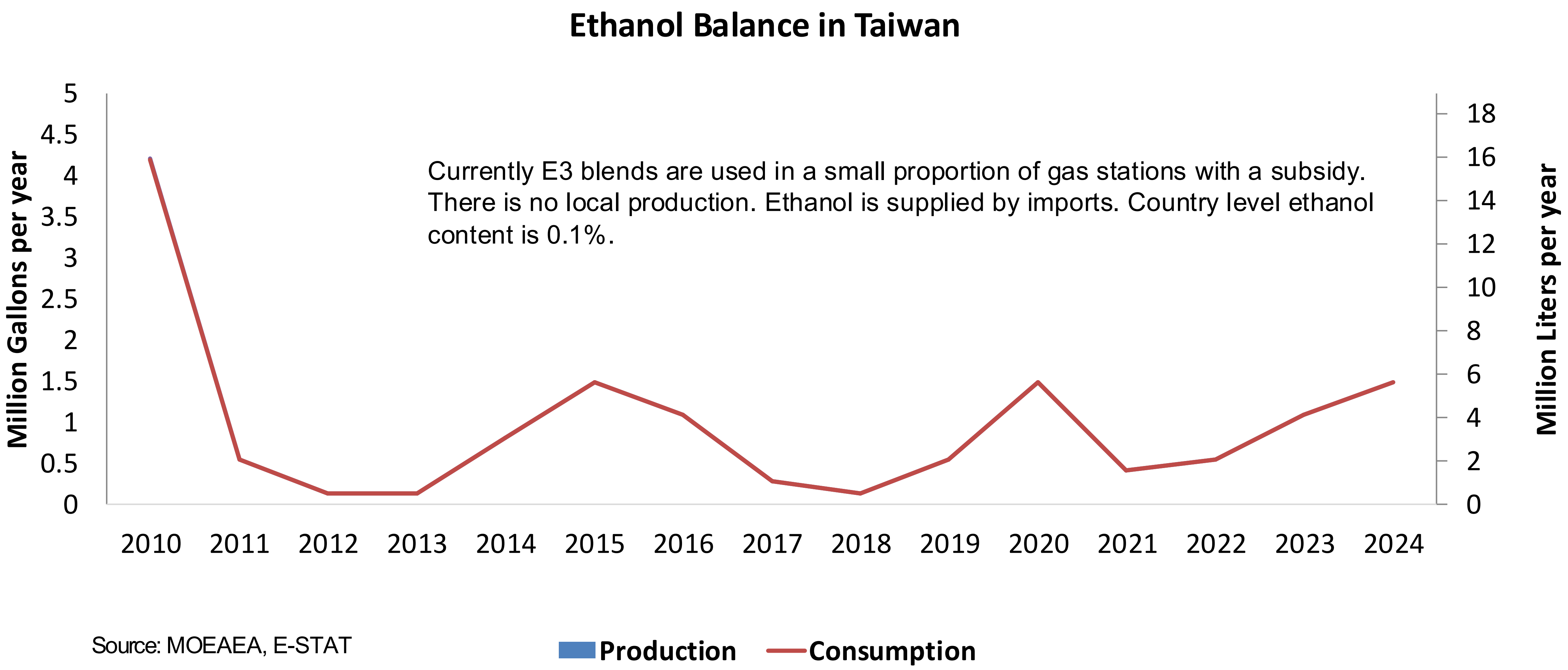
Source: MOEAEA, E-STAT

Gasoline Exports in Taiwan



Source: MOEAEA, E-STAT

Ethanol Balance in Taiwan



Source: MOEAEA, E-STAT

Gasoline Quality in Taiwan

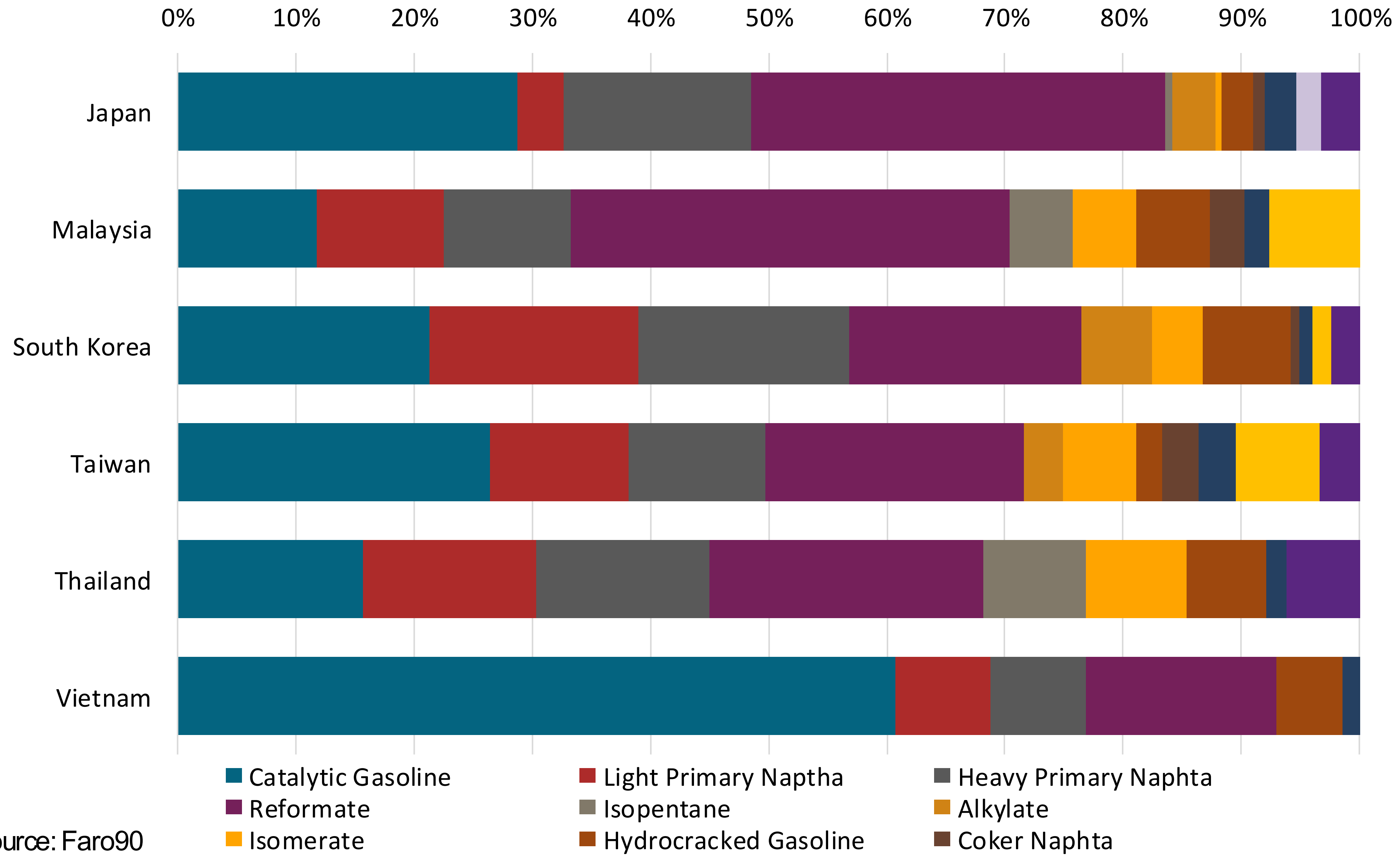
Name	Mobile Pollution Source Fuel Composition Control Standard Regulation					
Implementation Year	2024					
Applicability	Nationwide					
Grade	92 RON		95 RON (Standard)		98 RON (Premium)	
	Min	Max	Min	Max	Min	Max
RON	92.0		95.0		98.0	
MON						
AKI						
Benzene (%vol)		0.8		0.8		0.8
Aromatics (% vol)		35.0		35.0		35.0
Olefins (%vol)		18.0		18.0		18.0
Lead (g/L)						
Manganese (mg/L)						
Sulfur (ppm)		10		10		10
Oxygen (%v)		3.2		3.2		3.2
Ethanol (%vol)		10.0		10.0		10.0
MTBE (%vol)						
RVP (kPa)	45.0	66.9	45.0	66.9	45.0	66.9

Source: MOEAEA, MOENV, Formosa Petrochemical

Gasoline Component Blending in Asia

Gasoline is a blend of a base gasoline and other components. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Asia are:

- Catalytic Gasoline
- Light Primary Naptha
- Heavy Primary Naptha
- Reformate
- Isopentane
- Alkylate
- Isomerate
- Hydrocracked Gasoline
- Coker Naphta
- n-Butane
- MTBE
- ETBE
- TAME
- Aromatics



Source: Faro90

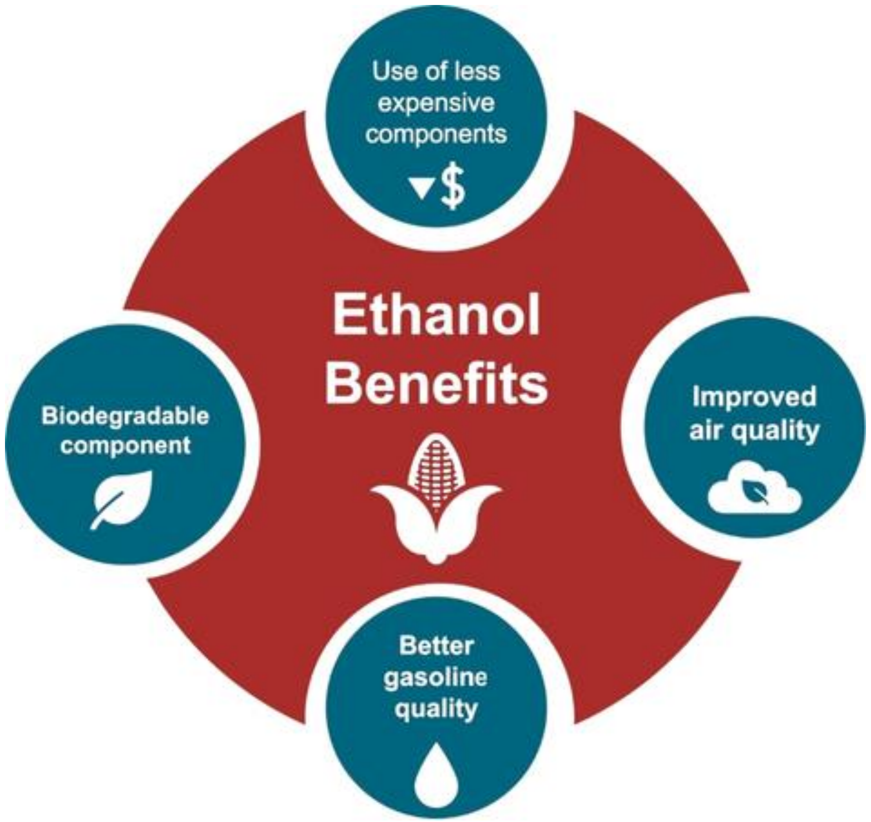
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component.The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives.To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

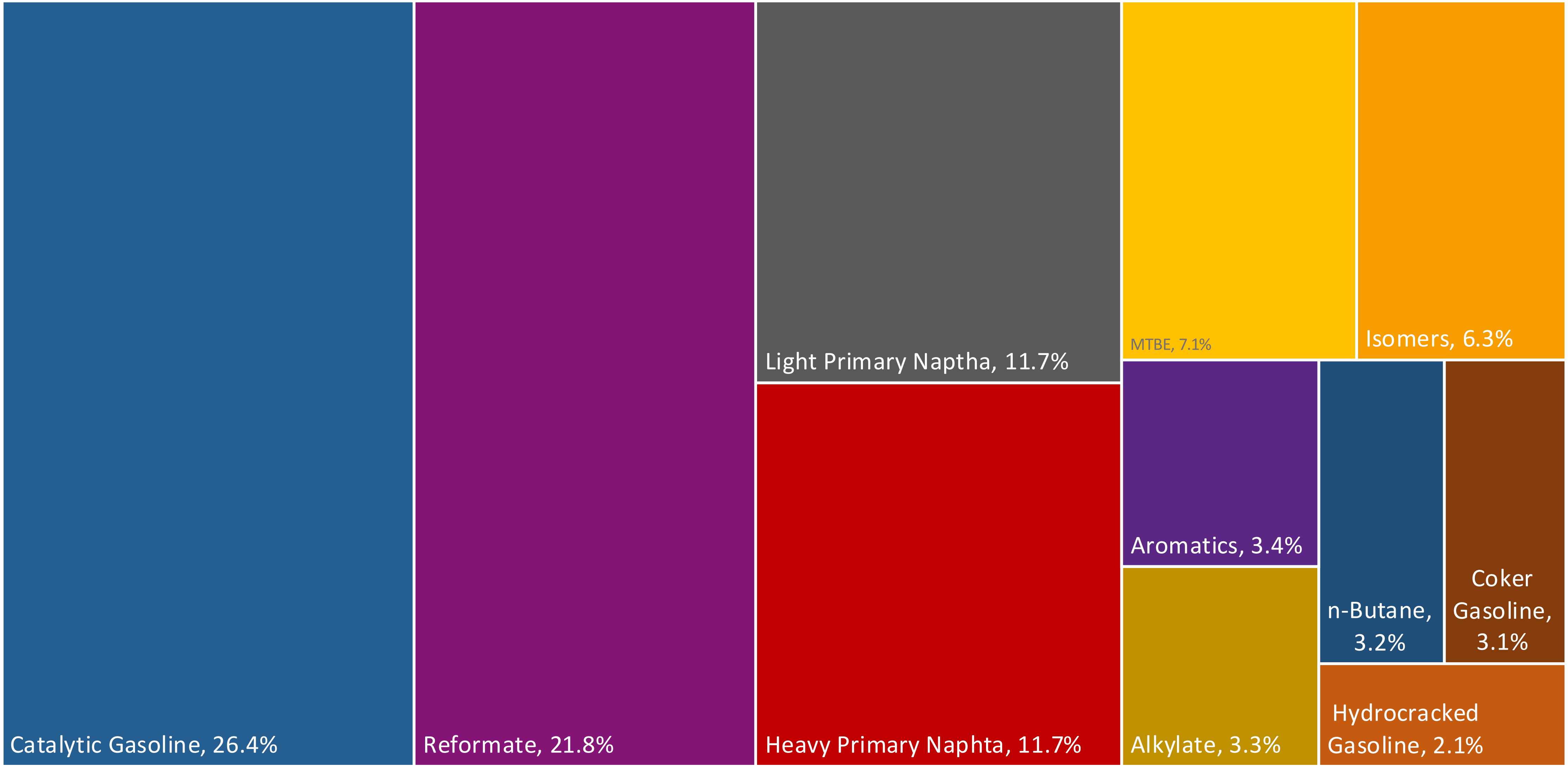
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

The blending model uses gasoline component spot average prices 2024 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.

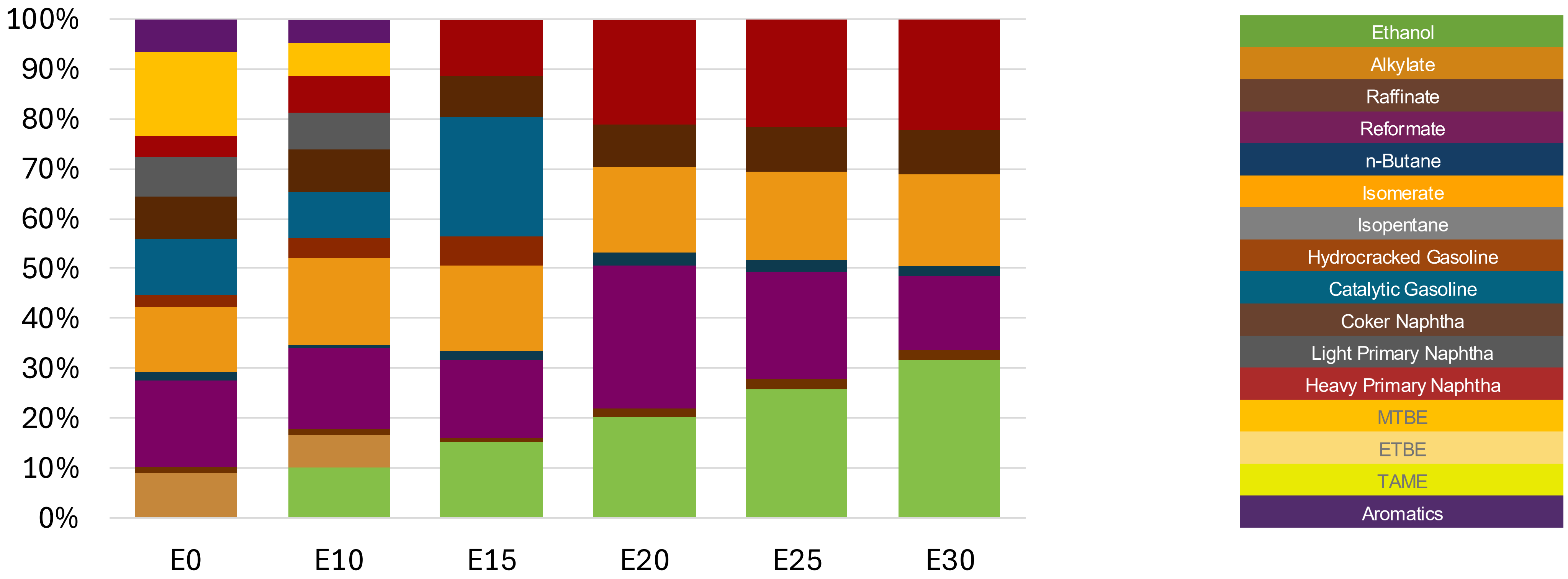


Taiwan Available Blending Components



Source: Faro90

Taiwan – Gasoline 95 – Constant Octane

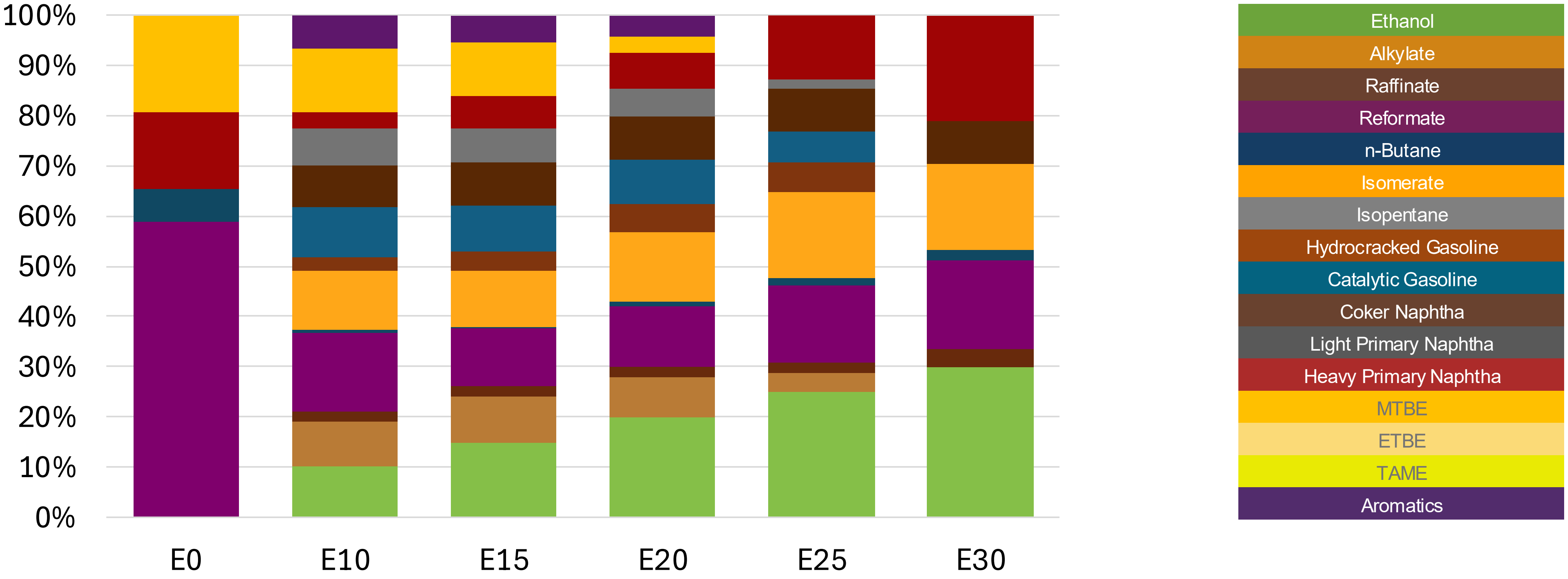


Octane (RON)	95.0	95.0	95.0	95.8	95.0	95.0
Price (US$/gal)	$ 2.388	$ 2.307	$ 2.223	$ 2.149	$ 2.061	$ 1.995

Average CIF 2024 Prices

Source: Faro90

Taiwan - Gasoline 98 - Constant Octane

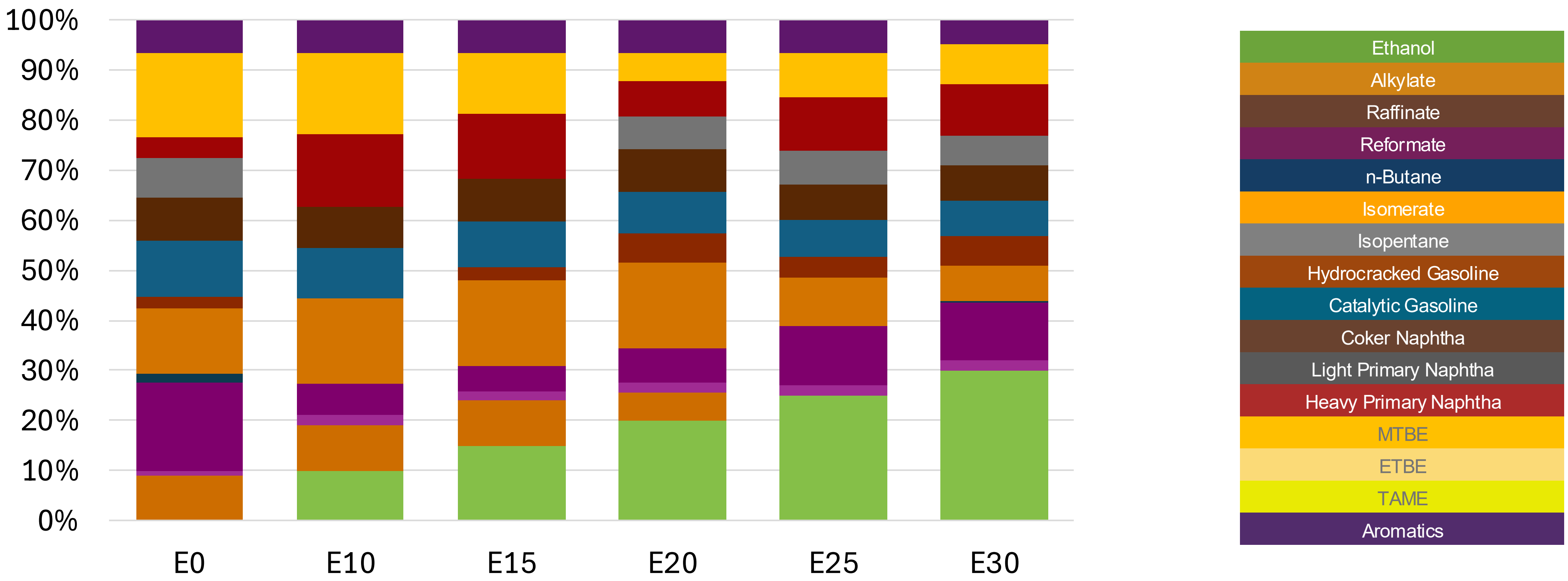


Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.6
Price (US$/gal)	$ 2.445	$ 2.418	$ 2.346	$ 2.305	$ 2.210	$ 2.117

Average CIF 2024 Prices

Source: Faro90

Taiwan – Gasoline 95 – Increased Octane

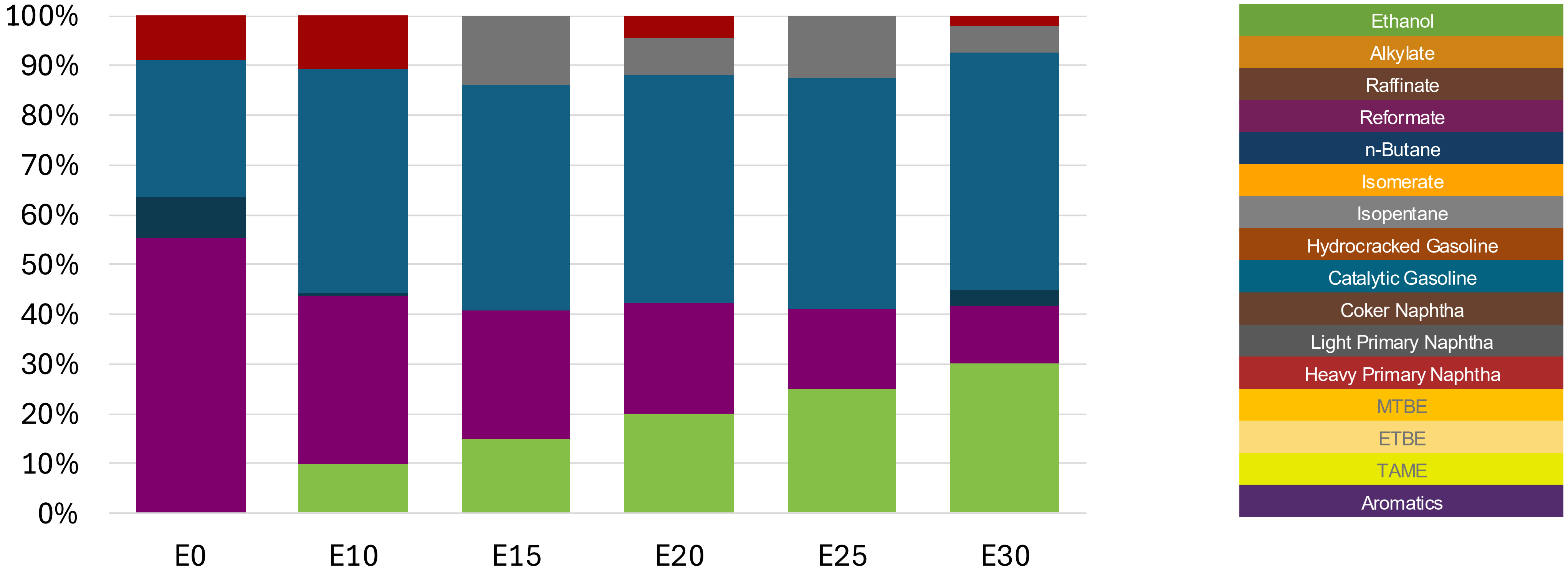


Octane (RON)	95.0	96.4	97.4	98.1	100.6	102.0
Price (US$/gal)	$ 2.388	$ 2.388	$ 2.388	$ 2.388	$ 2.388	$ 2.388

Average CIF 2024 Prices

Source: Faro90

Taiwan – Gasoline 98 – Increased Octane



Octane (RON)	98.0	99.8	99.9	101.3	102.6	103.9
Price (US$/gal)	$ 2.445	$ 2.445	$ 2.445	$ 2.445	$ 2.445	$ 2.445

Average CIF 2024 Prices

Source: Faro90

Vehicle Emission Impact for Ethanol Gasoline Blending

The model used in this analysis takes as a reference the **International Vehicle Emissions Model (IVE).**
The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

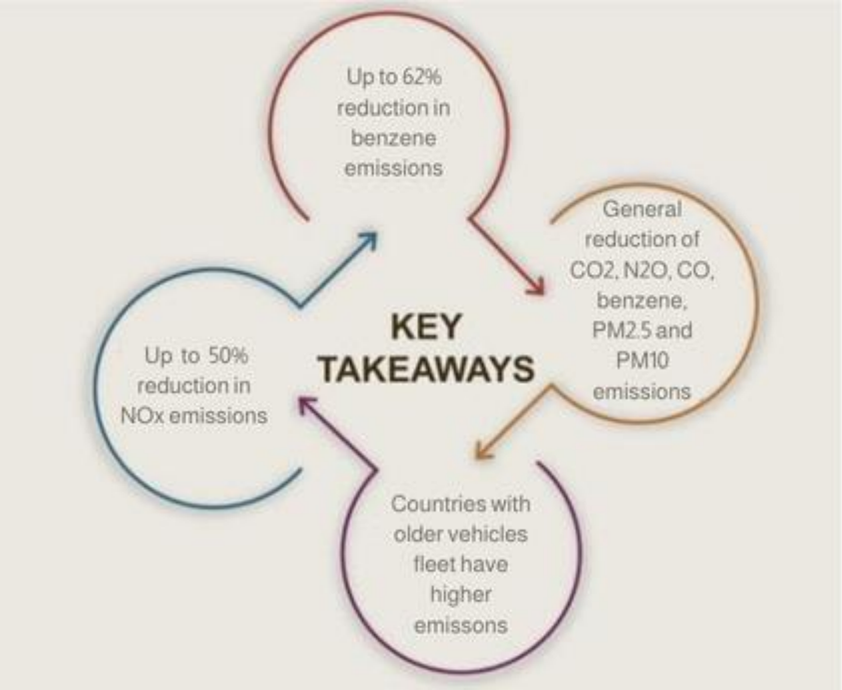
- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

**Source: Bureau of transportation statistics.*

Main Results



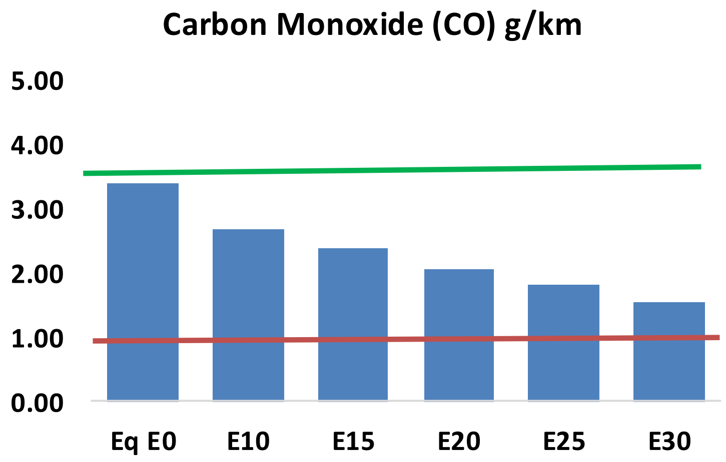
Gasoline Vehicle Fleet - Taiwan

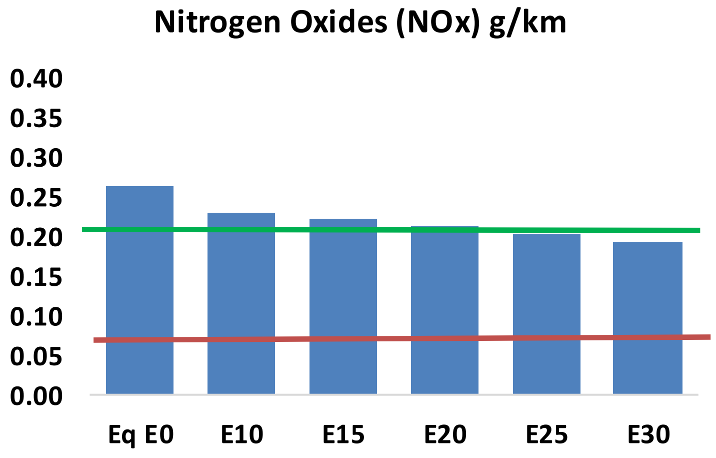
Type	Weight	Motor	Escape	Recover	Age	No. Vehicles
Auto/Sml Truck	Light	Carburator	3-Ways	PCV Tank	>8 years	9,555
		Single FI	3-Ways		>8 years	277,656
		Multi FI	3-Ways		>8 years	781,852
			Euro III		>8 years	713,122
			Euro IV		>8 years	943,717
			Euro V		>8 years	1,415,819
			Euro 6		>8 years	1,156,342
					4-8 years	981,907
					<4 years	960,692
			Hybrid		All	143,305
		Electric			All	48,801
	Medium	Carburator	3-Ways	PCV Tank	>8 years	17,020
		Single FI	3-Ways		>8 years	67,268
		Multi FI	3-Ways		>8 years	226,812
			Euro III		>8 years	117,564
			Euro IV		>8 years	177,121
			Euro V		>8 years	88,574
			Euro 6		>8 years	99,272
					4-8 years	94,150
					<4 years	109,767
			Hybrid		All	3,244
		Electric			All	974
Sml Engines	Light	4-Cycle	None	PCV	>8 years	884,870
			Euro III		>8 years	2,061,596
			Euro IV		>8 years	3,370,098
			Euro V		>8 years	3,408,923
			Euro 6		>8 years	203,066
					4-8 years	1,706,126
					<4 years	2,057,361
			Hybrid		All	0
		Electric			All	808,708
Truck	Heavy	Carburator	3-Ways	PCV	>8 years	16
		Fuel-Injection	3-Ways		>8 years	10,919
			Euro III		>8 years	16,174
			Euro IV		>8 years	53,747
			Euro V		>8 years	30,133
			Euro 6		>8 years	61,123
					4-8 years	57,191
					<4 years	50,710
			Hybrid		All	11,543
		Electric			All	12,195

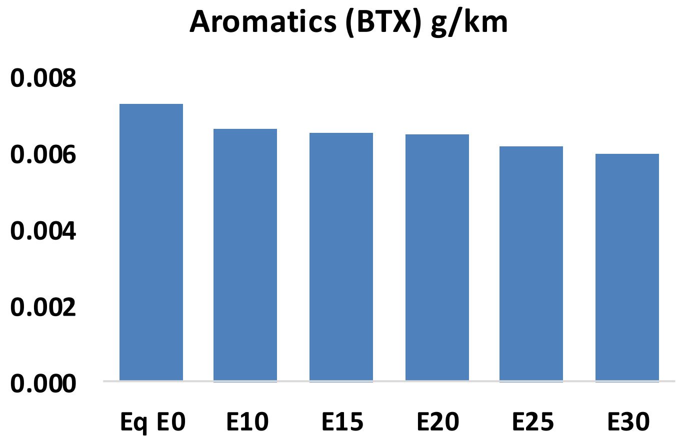
Vehicle Fleet: 23,239,031
Gasoline Vehicle Fleet: 20,069,210

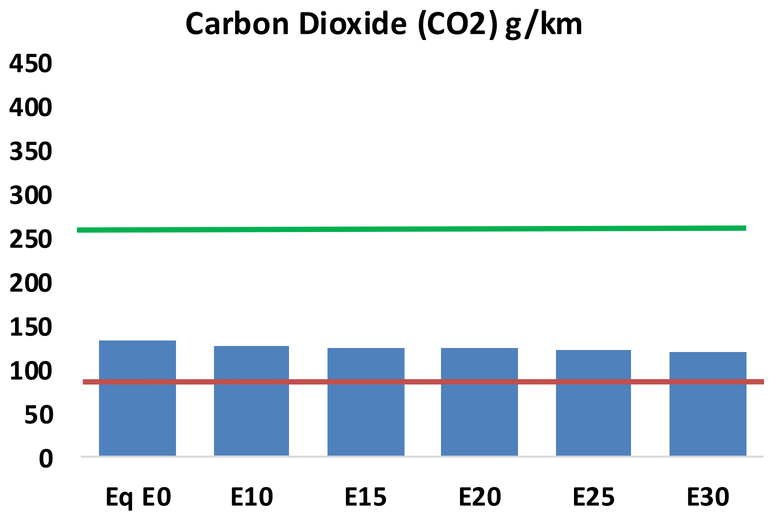
Source: DATA-GOV.tw, CIECData

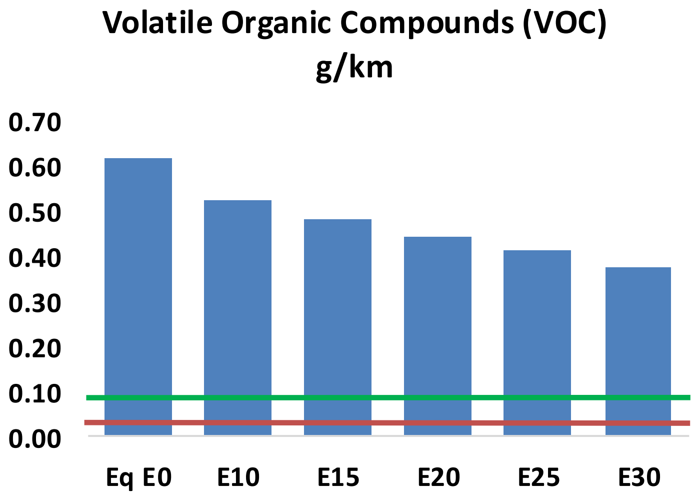
Taiwan – Vehicular Emissions

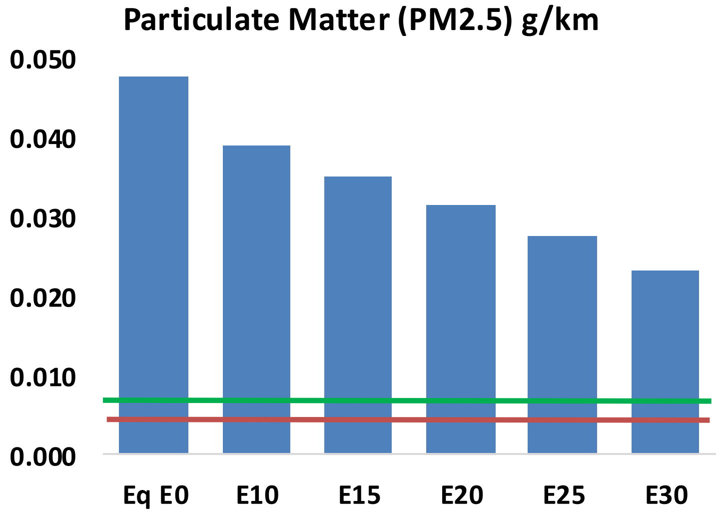












Source: Faro90

Taiwan – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	1,603,124	1,435,644	1,268,164	1,122,290	971,520	854,387	729,007	-21%	-39%
CO2	62,635,536	61,024,908	59,503,759	58,307,420	57,715,425	57,451,104	56,392,115	-5%	-8%
VOC	291,022	268,550	246,078	227,644	209,084	194,810	175,771	-15%	-28%
NOx	124,311	114,781	108,287	105,003	99,686	95,530	90,629	-13%	-20%
SOx	686	634	582	538	496	451	403	-15%	-28%
NH3	26,238	25,978	25,718	25,840	26,131	26,335	26,505	-2%	0%
N2O	766	762	759	763	779	787	796	-1%	2%
CH4	79,981	79,981	79,981	81,180	82,940	84,459	85,899	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	2,061	1,917	1,773	1,659	1,545	1,457	1,339	-14%	-25%
Acetaldehyde	6,412	7,359	10,797	16,443	22,402	25,598	30,247	68%	249%
Formaldehyde	25,834	25,116	29,278	34,379	36,017	39,844	43,375	13%	39%
Benzene	3,435	3,301	3,128	3,078	3,053	2,920	2,825	-9%	-11%
PM2.5	4,384	3,986	3,587	3,236	2,893	2,525	2,140	-18%	-34%
PM10	18,103	16,457	14,812	13,363	11,946	10,427	8,838	-18%	-34%

Source: Faro90

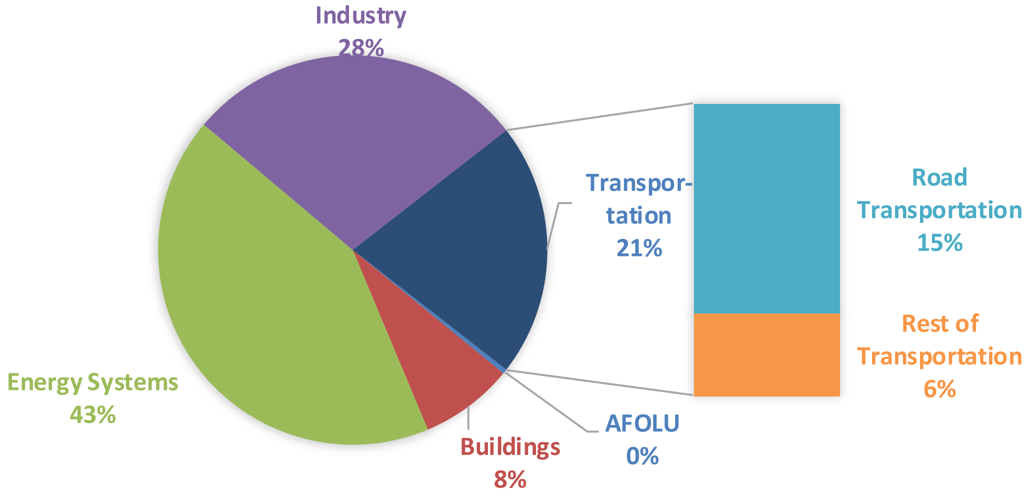
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

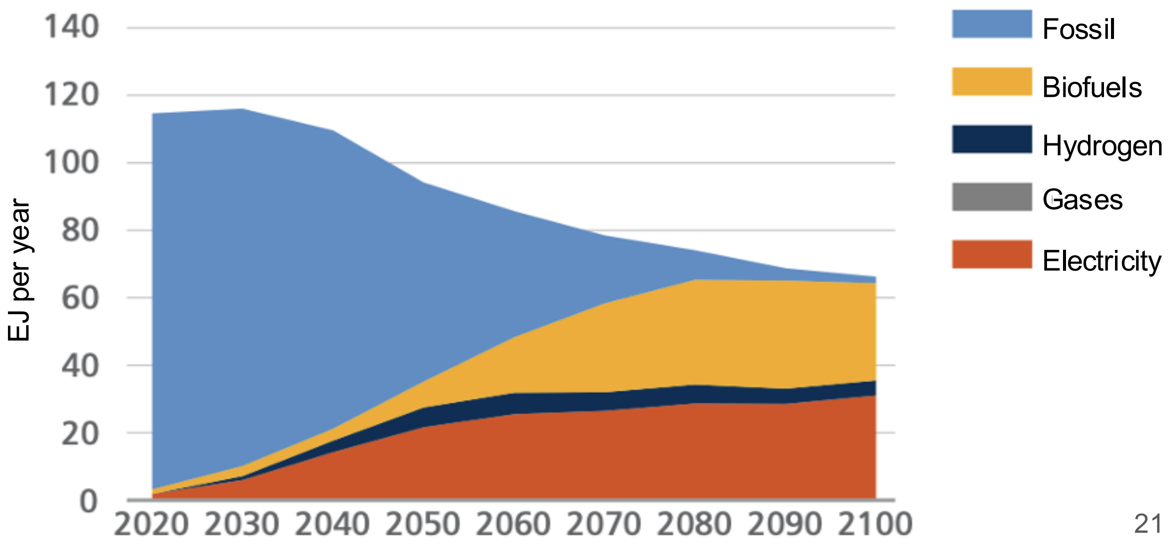
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE).** Results are compared to the current country targets for 2035. Current 2024 country emission are those published by **IPCC**.

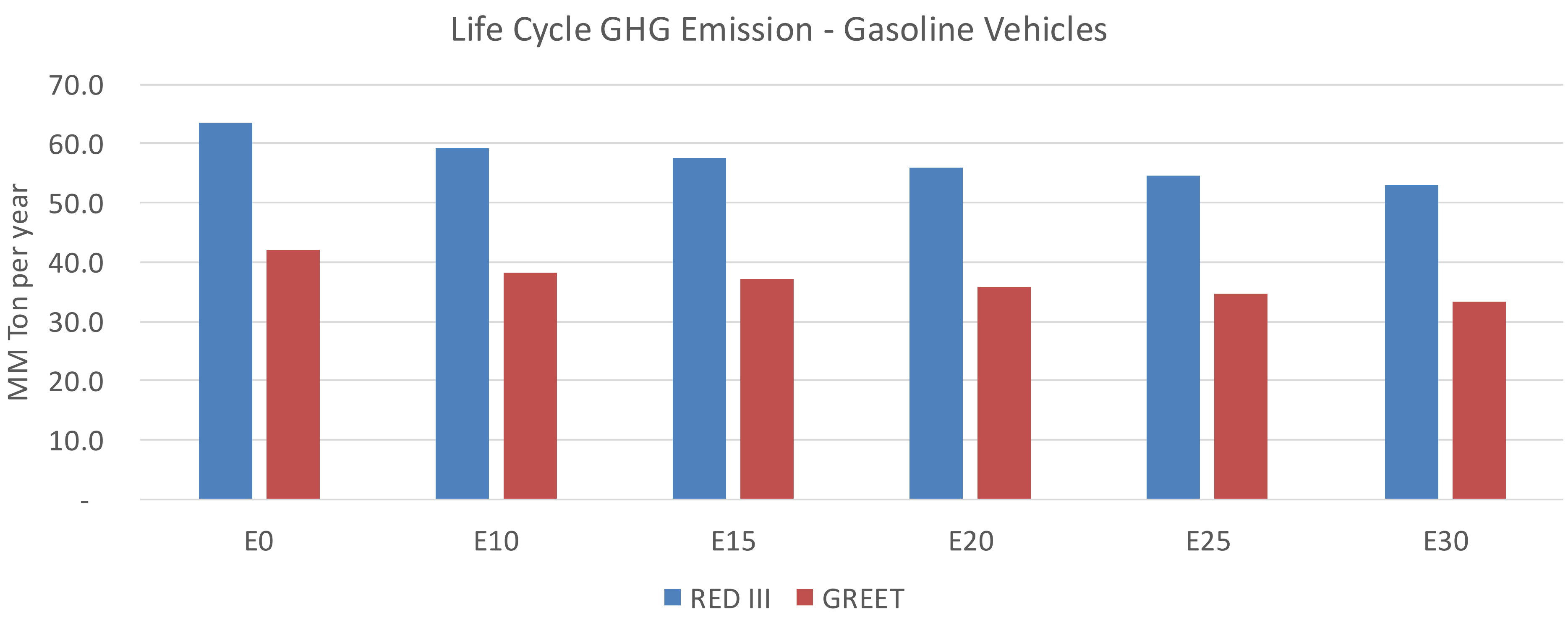
Current GHG Global Emissions Distribution



IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



Taiwan – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	291.7
Target @ 2035 (MMT/year)	204.2
Delta	87.5

vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	8.9%	11.8%	14.8%	17.2%	20.5%
RED II/III	0.0%	7.0%	9.5%	12.0%	14.1%	16.7%

Target	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	4.3%	5.7%	7.1%	8.2%	9.8%
RED II/III	0.0%	5.1%	6.9%	8.7%	10.2%	12.2%

Source: Greet, RED II/III, climateactiontracker.org, F90